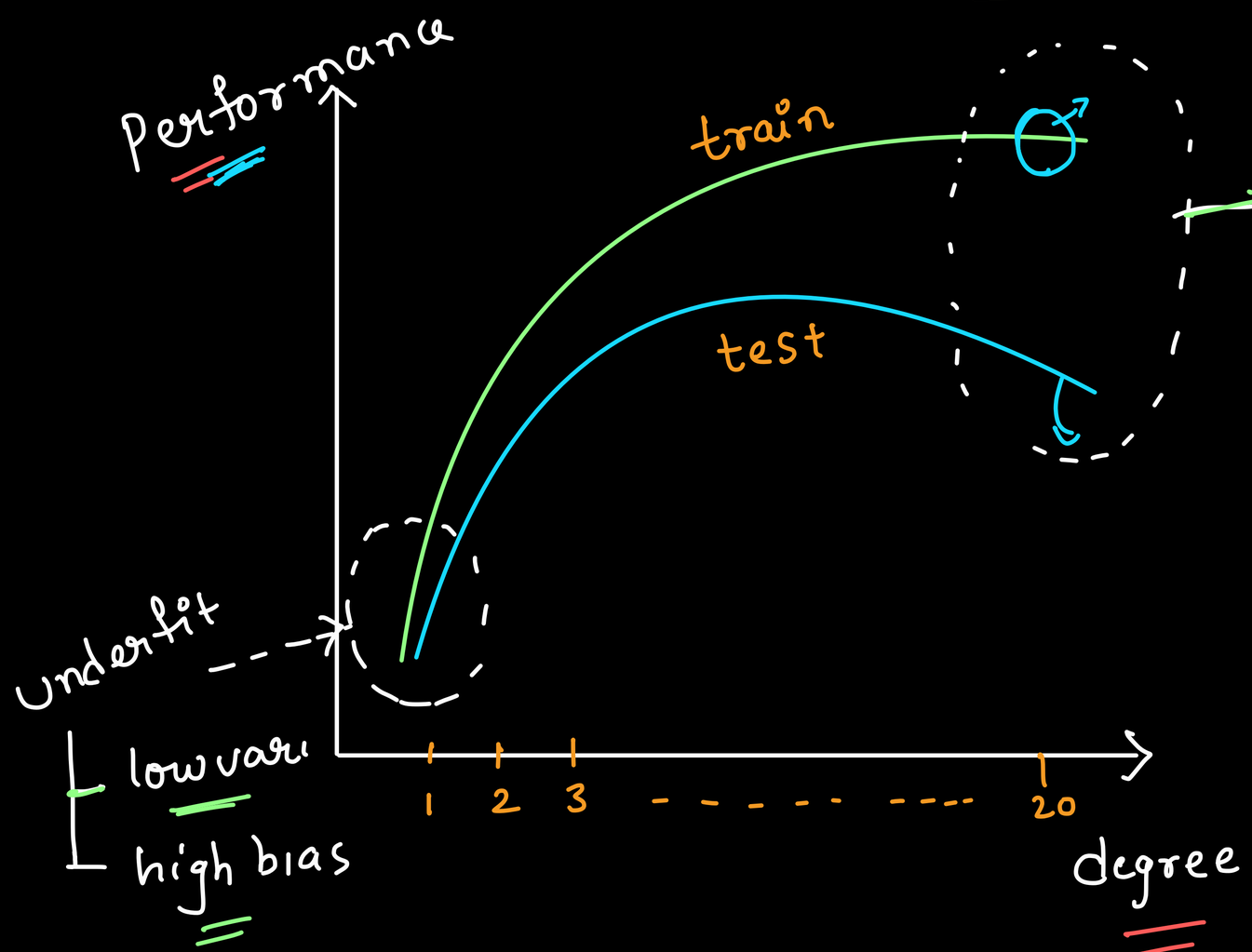
Agenda

* Regularization Code
* Hyperparameter Tuning
* Cross Validation
* K- Fold Cross Validation
* Logistic Regression ✓✓

Assignments

Made with Goodnotes

Recap



overfit

high variance

low bias



Made with Goodnotes

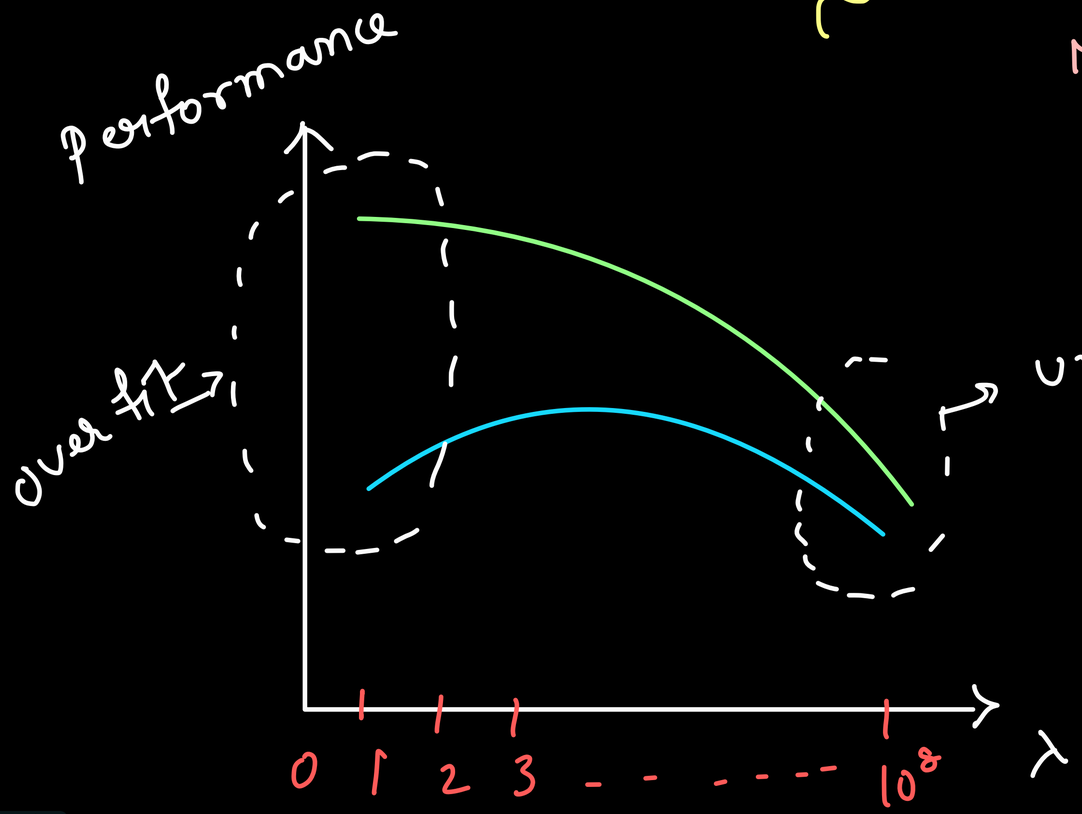
Regularisation:

New Loss function

$$= \frac{1}{n}\sum_{i=1}^{n}\left(y^{(i)} - \hat{y}^{(i)}\right)^2 + \lambda \sum_{i=1}^{d} w_j^2$$

MSE

Regularisation



Reg. parameter

$\lambda \geq 0$

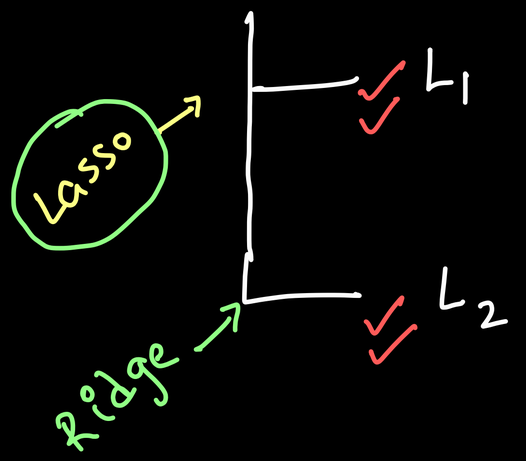
train , test

$\lambda = 0$ (high , low)

$\lambda = 10^8$ (low, low)

Made with Goodnotes

Types of Regularisation



L_1 Regularisation → $|w_j|$ → lot of weights become 0.

$$\frac{\partial |w_j|}{\partial w_j} = \begin{cases} +1 & w_j > 0 \\ -1 & w_j < 0 \\ \simeq 0 & w_j = 0 \end{cases}$$

→ Property Sparsity

hack

L_2 Regularisation → w_j^2

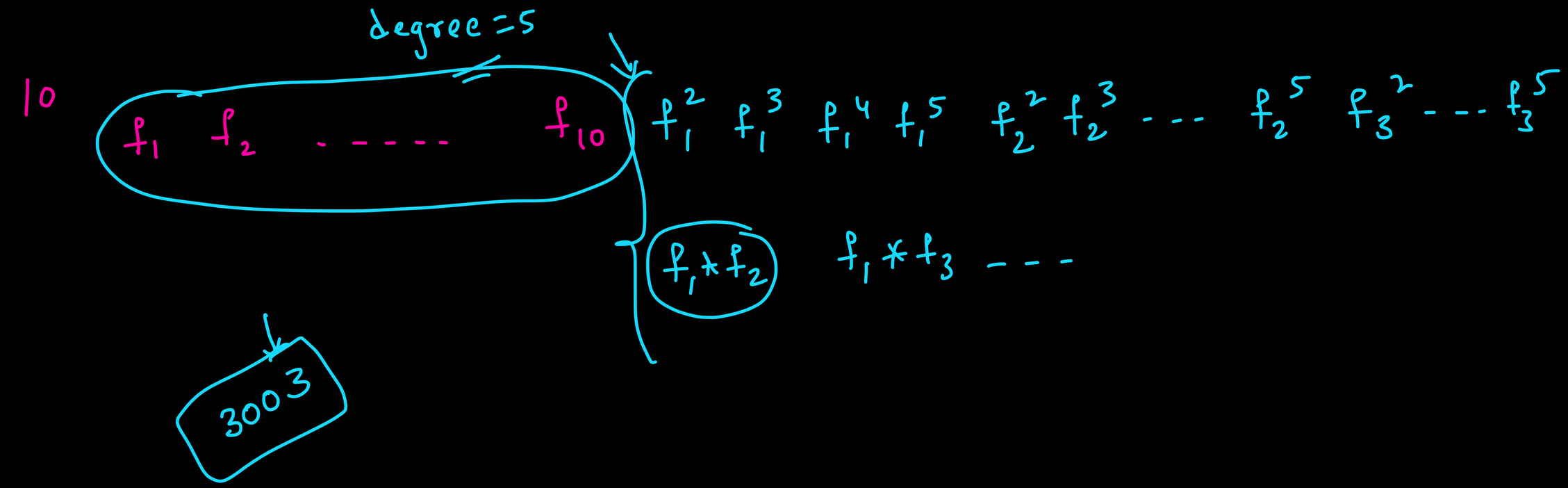
↳ no sparsity

→ weights will be closer to 0 but $\neq 0$ ✓

Elastic Net

$$MSE + \lambda_1 \sum_{i=1}^{d} |w_j| + \lambda_2 \sum_{i=1}^{d} w_j^2$$

Made with Goodnotes



Made with Goodnotes

Hyper parameter Tuning

$\hat{y} = \vec{w}^T \vec{x} + w_0$

✓ Parameters : Coef / weights

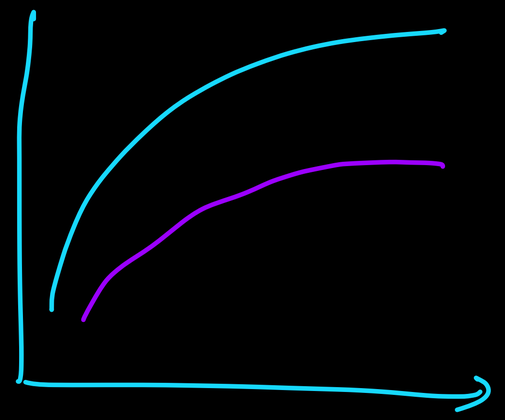
Hyper parameters : α, λ, η — hyper parameters

How do we find the best value of hyper parameter?

Hyper parameter tuning

Made with Goodnotes

λ



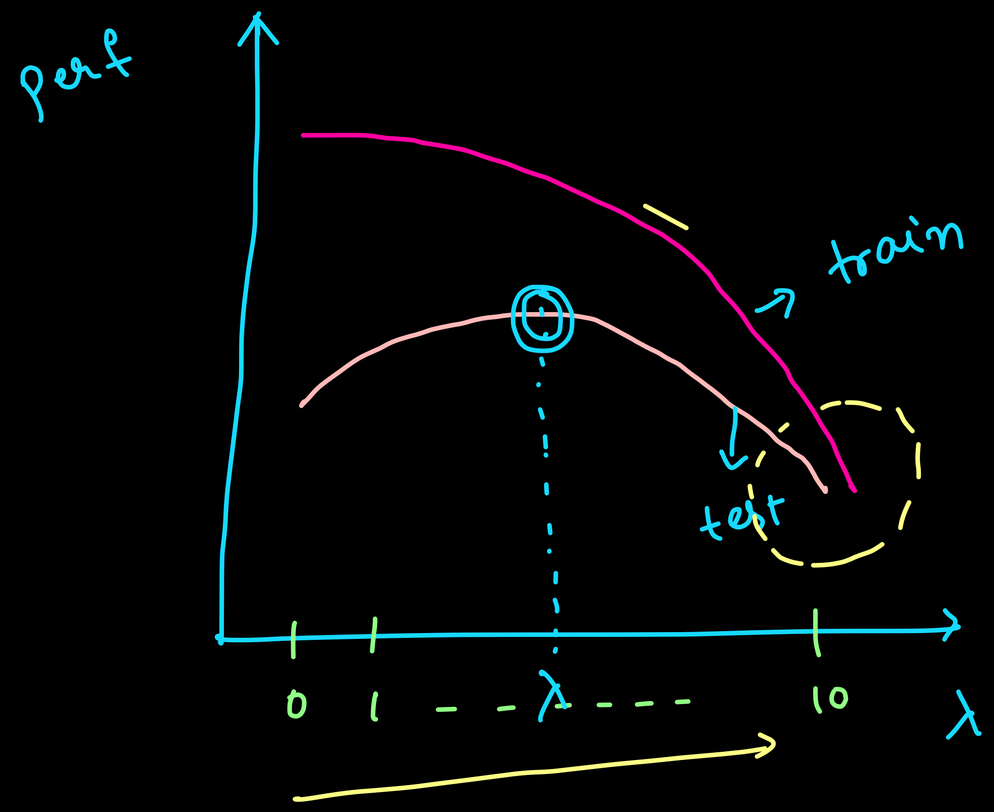
```
train_score = []
test_score = []

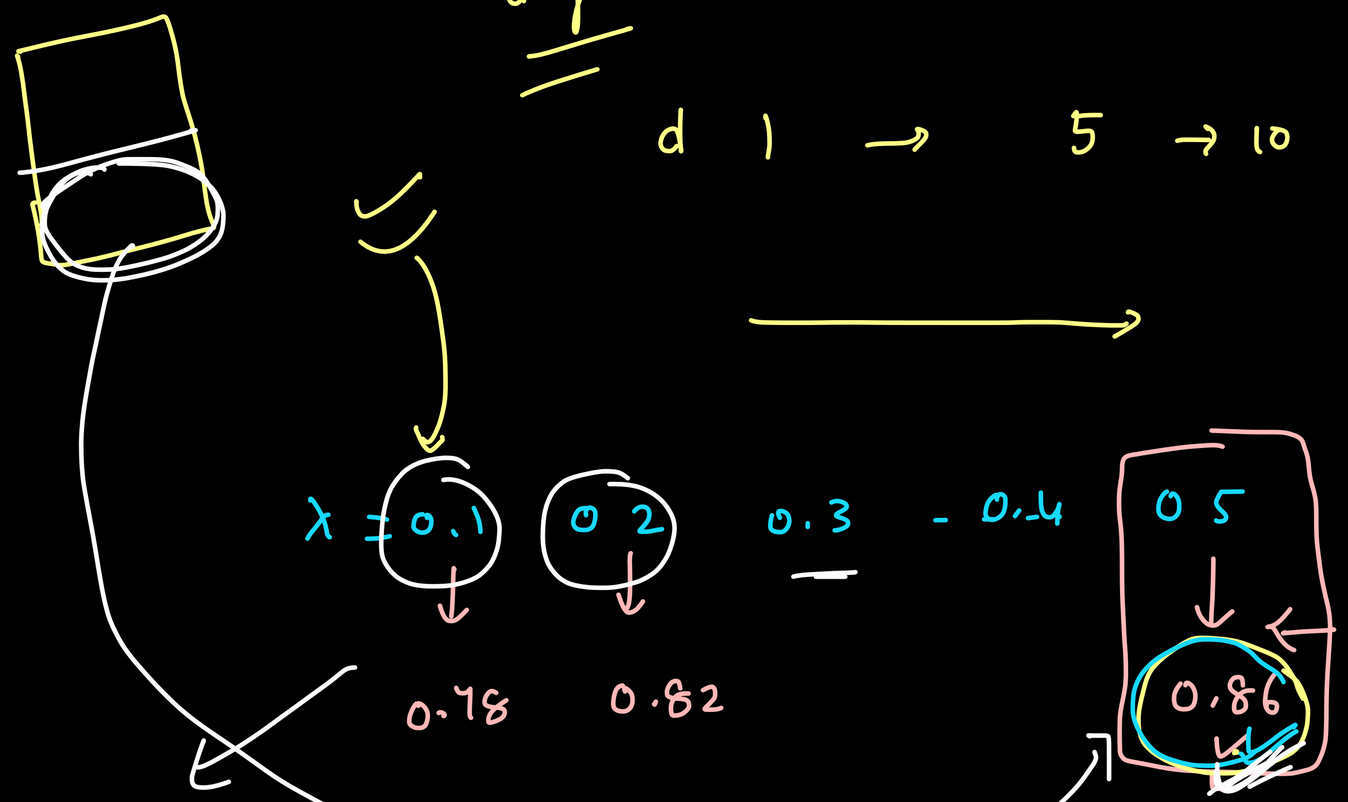
for λ in range(0.1, 10, 0.1)
    → model = Lasso(λ)
    → model.fit()        // train the model

    → { test_per
    → { train_per

    → append
```

Made with Goodnotes



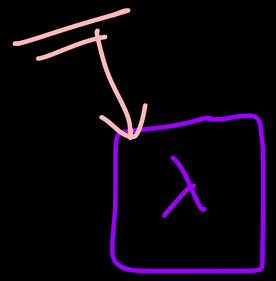


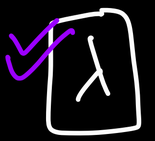
Future

Expectation of the model

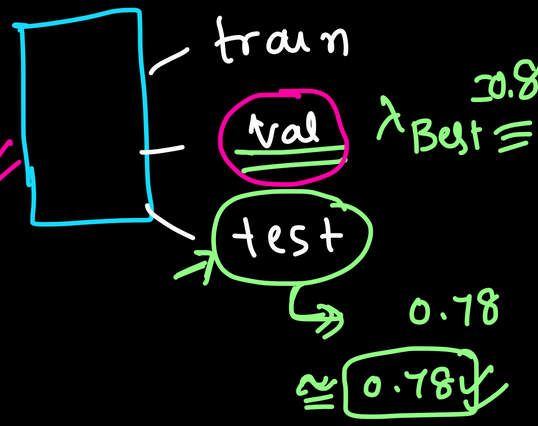
≃ 0.80

Used test data multiple times

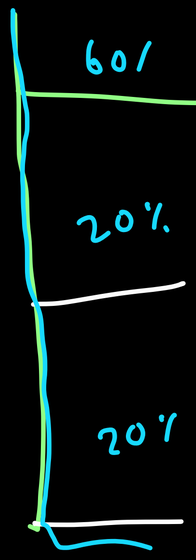




= 0.1 0.2 ---- 0.5



Data (80%, 10, 10)



D_{train} ⟶ train the model

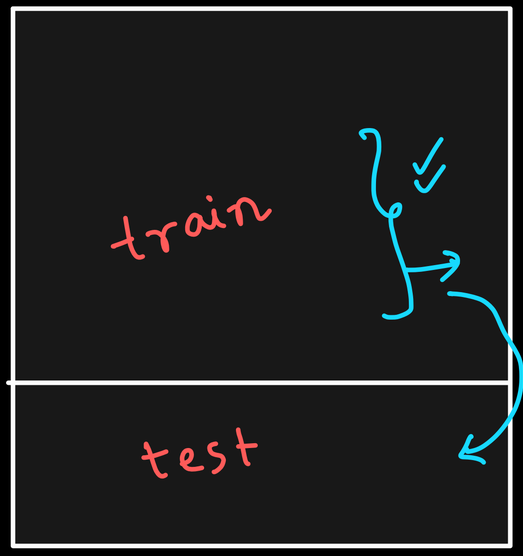
D_{cv} (or) D_{val} ⟶ used to do the hi-P tuning (λ, d_ ->

≡

D_{test} ⟶ to report the final performance / score

Made with Goodnotes

Cross validation

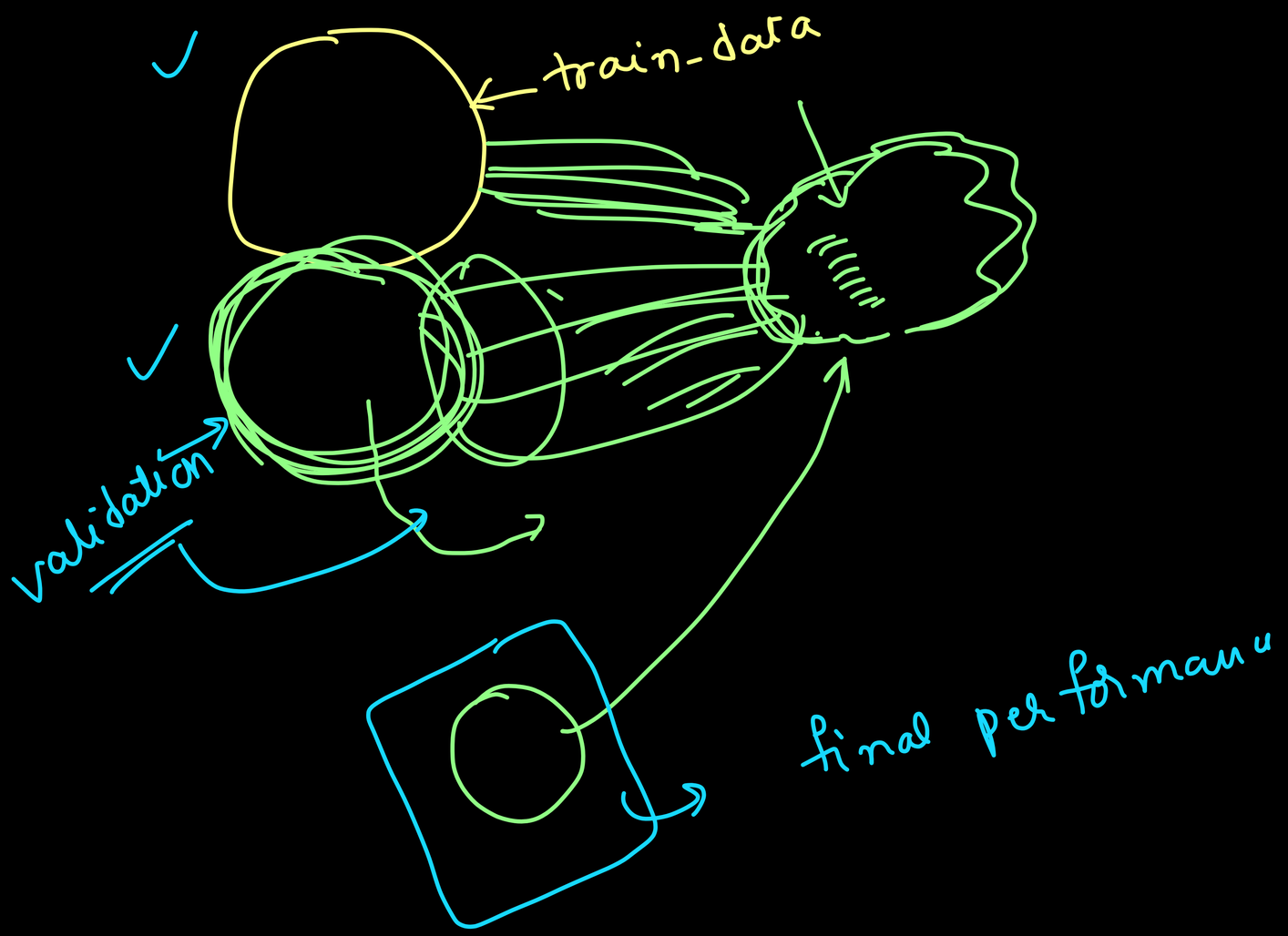


Test data → R^2-score → on test data

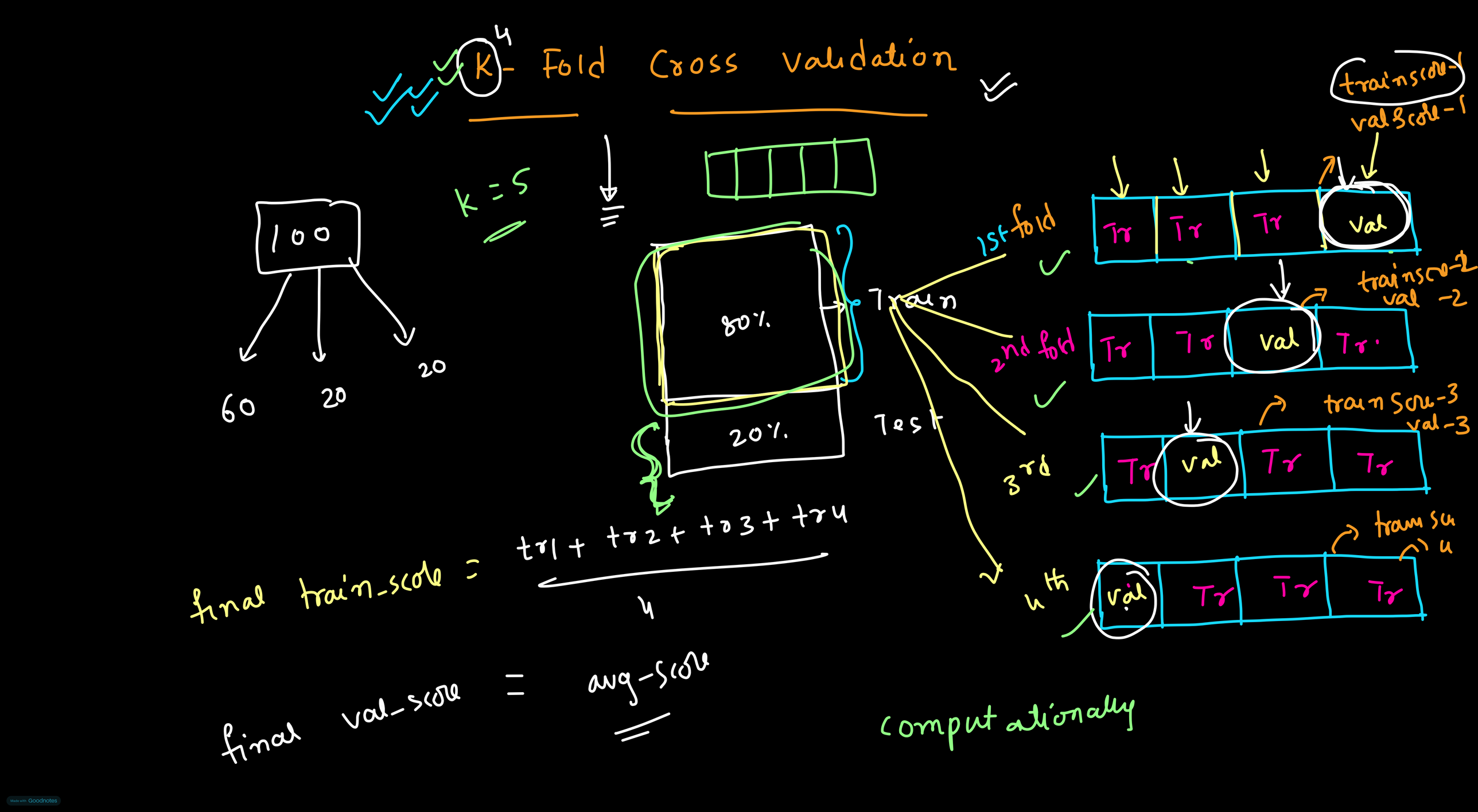
→ evaluate how good the model is

10:06 → 10.11

Made with Goodnotes


train-data
validation
final performance

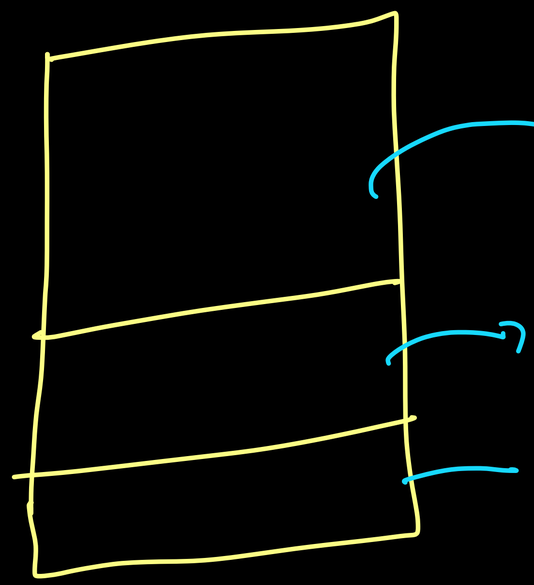
K-Fold Cross Validation




1
100
Kf.s
1 fold
1
91 92 93 --100
val
train
80
90
val
trai

1 m

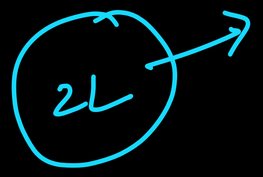
80, 10, 10
60 20 20



800K

6L





2L

2L

Made with Goodnotes